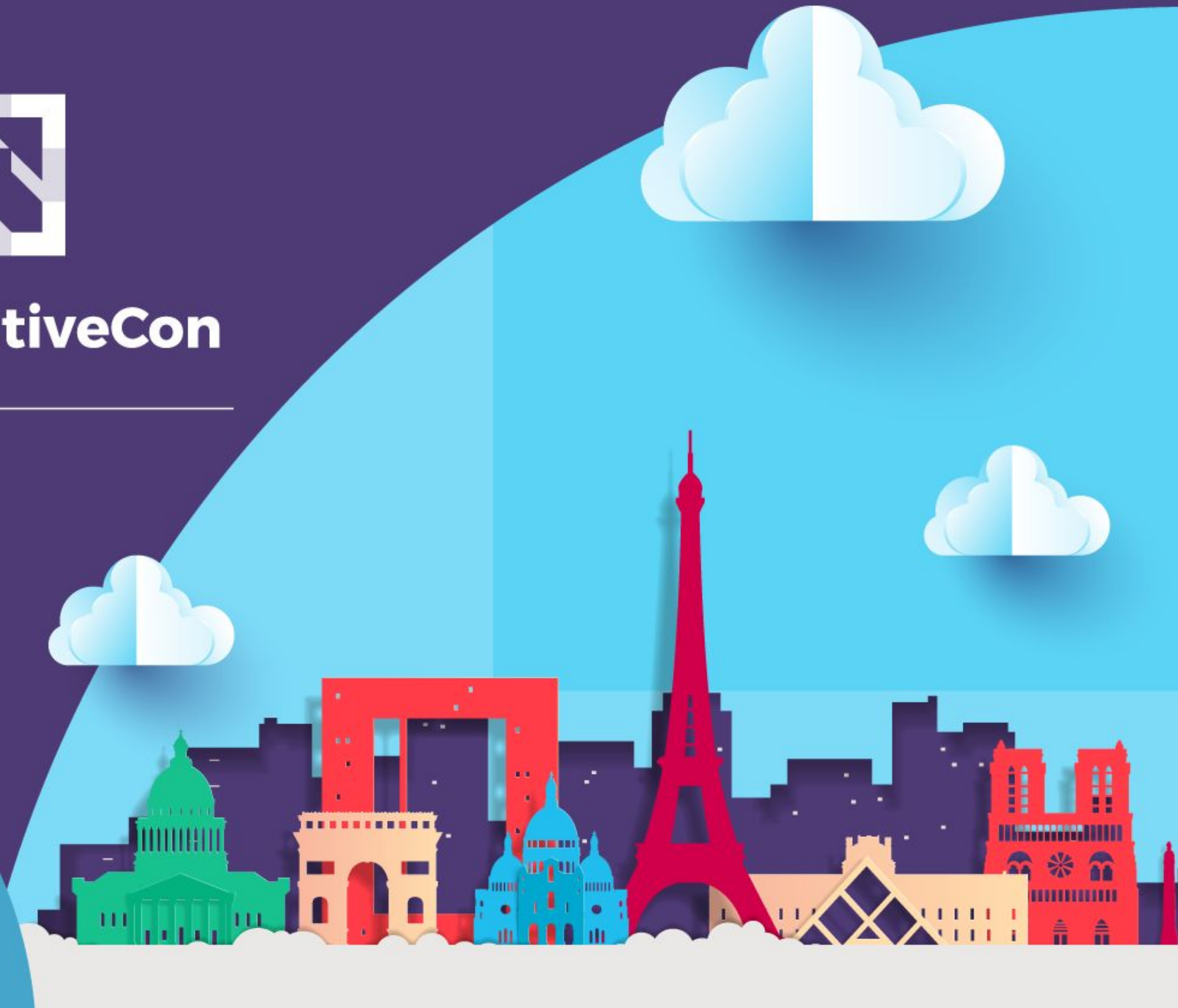
KubeCon | CloudNativeCon

Europe 2024



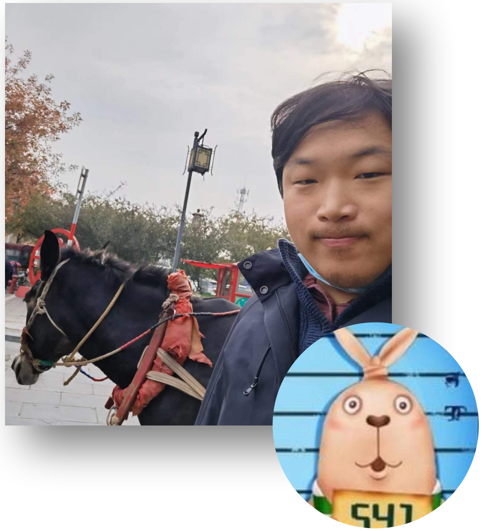
How to deploy an AI-Optimized K8s cluster with Kubespray

Mohamed Zaian & Kay Yan

About us

Mohamed Zaian

Senior Systems Engineer
New Work SE, Germany
Github ID: mzaian
Kubespray Maintainer



Kay Yan

Principal Engineer
DaoCloud, China
Github ID: yankay
Kubespray, Containerd/Nerdctl Maintainer

Deploy Kubernetes >500 various environment have been delivered in since 2016

Agenda

1. What is Kubespray?
2. Features & Best Practices
3. Build an AI-Optimized Cluster
4. Demo
5. Community
6. QA

Kubespray

KUBESPRAY

What is Kubespray

what is kubespray

GPT-4-32k Poe

Kubespray is an open-source project that provides tooling to deploy and manage production-ready Kubernetes clusters. It uses Ansible, which is a powerful automation tool, to handle the installation of Kubernetes on a variety of host infrastructures, including public cloud providers like AWS, GCP, and Azure, as well as private data centers and bare-metal servers.

Ref: Dragon Ball - Akira Toriyama 🙏

kubernetes

ANSIBLE

When Using Kubespray

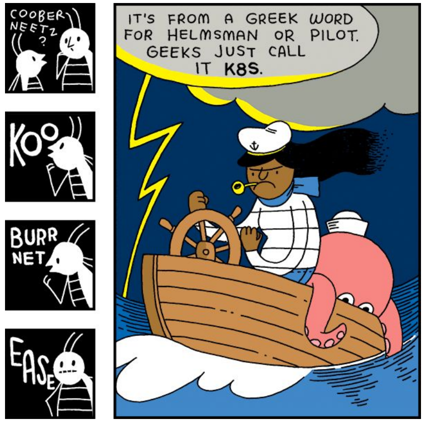
BootStrap
K8S SIG
kubeadm

Provisioner
K8S SIG
kops
Bare Metal
Production
Easy

Public Cloud
Amazon EKS
KUBERNETES ENGINE
Azure Kubernetes Service (AKS)

Developer
K8S SIG
minikube
kind

Commercial
RED HAT OPENSHIFT Container Platform
VMware Tanzu
DaoCloud Enterprise



Main Features

- **Flexible** Can be used with **Cloud Environments (`terraform`) or Bare Metal (`ansible`).**
- **Highly availability** cluster(s) e.g control plane and etcd.
- **Configuration Options** (various aspects choice of CRI, CNI ..etc).
- Supports most popular **Linux distributions.**
- **Continuous integration tests.**

Options you can choose!

Cloud	Supported Linux	CRI	CNI	CSI	Others
AWS	Flatcar Container Linux	Containerd	Calico	cephfs-provisioner	Coredns
Google Cloud	Ubuntu 20.04, 22.04	Docker *	Cilium	rbd-provisioner	Metallb
Equinix	CentOS/RHEL/Oracle 7, 8, 9	CRI-O	cni-plugins (MacVlan…)	aws-ebs-csi-plugin	Ingress-nginx
Huawei Cloud	Alma/Rocky Linux 8, 9	Crun	Multus	azure-csi-plugin	Kube-vip
Upcloud	Fedora 37, 38; CoreOS	Gvisor	Flannel	cinder-csi-plugin	Cert-manager
VMWare vSphere	Debian 10,11,12	Kata	Cannel	gcp-pd-csi-plugin	ArgoCD
Openstack	OpenSUSE Leap 15.x/Tumbleweed	Youki	Weave	local-path-provisioner	Registry
Hetzner	Amazon Linux 2		Kube-OVN	local-volume-provisioner	Helm
Nif cloud	Kylin V10; UOS Linux; openEuler		Customize		Node-Feature-Discovery

Lifecycle of cluster operations

- Support full lifecycle of cluster operations
  - New cluster
  - Upgrade cluster or control plane component
  - Scaling a cluster
  - Node management e.g remove/add nodes
  - Reset cluster
  - Configuration management
- Backup and restore
  - etcd snapshots taken during upgrade.

Release Cycle in Kubespray

Kubernetes Support matrix

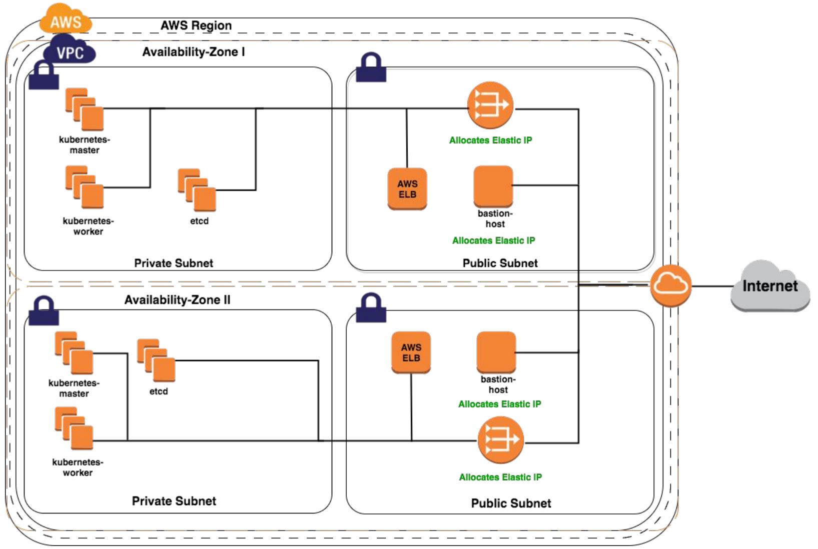
Kubespray	Kube Version (N-2)
master-branch	1.27 ~ **1.29**
2.24.1	1.26 ~ **1.28**
2.23.3	1.25 ~ **1.27**
2.22.2	1.24 ~ **1.26**

Release Cycle: A few weeks releases after the Kubernetes Release

Cloud Environment Deployment

Step1: Create VM/Net with Terraform

```
# Edit terraform config
# Apply with terraform
terraform init
terraform plan -no-color
terraform apply -no-color
```



Step2: Install k8s with Ansible

```
# Edit ansible config
# Apply with ansible
ansible-playbook … cluster.yml
```

Air-Gap Environment Deployment

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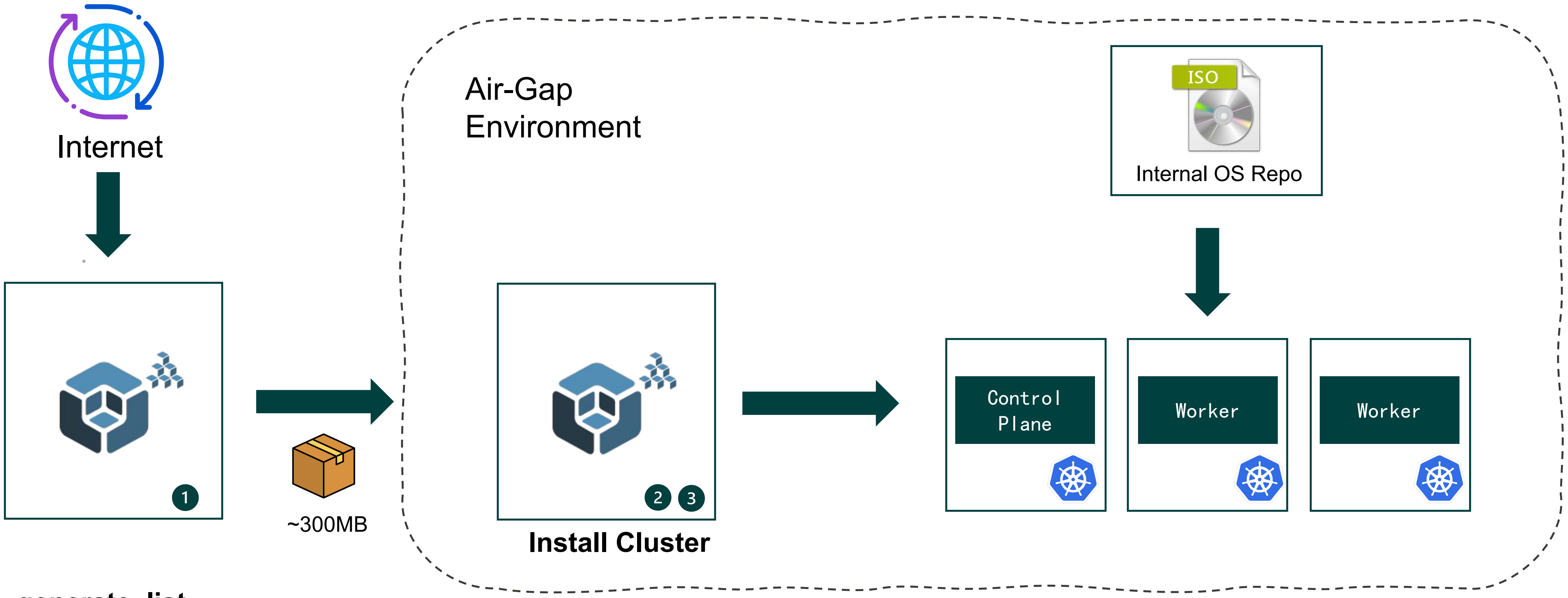


1 **generate_list**
2 **manage-offline-files**
3 **manage-offline-container-images**

Docker-CE is a Special Case!

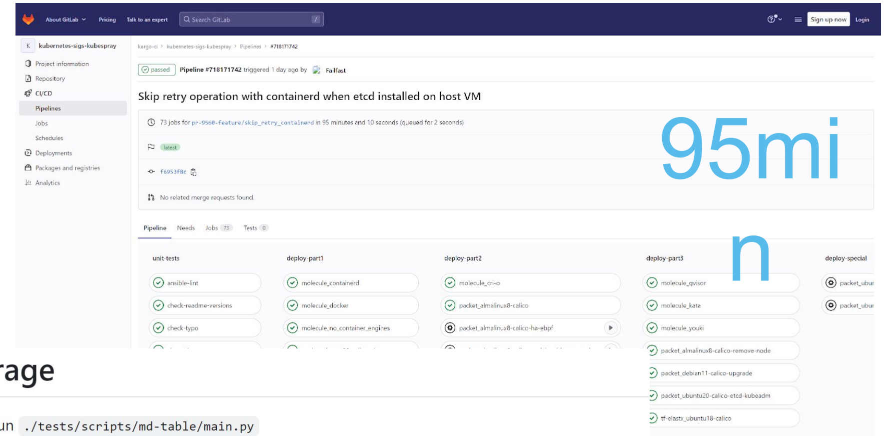
1. Not Support for Kubernetes in the Future.
2. Cray Dependency on Recent Version

Awesome CI Test

40+ test cases support:
- 13 operating systems
- 8 network plugins
- 30+ environments

CI strategies:
- All-in-one
- Separate roles
- HA
- Upgrade

CI Tips:
- Multi stage test cases
- Only run specific when coding
- Full automated



95min

CI test coverage

To generate this Matrix run `./tests/scripts/md-table/main.py`

containerd

OS / CNI	calico	cilium	custom_cni	flannel	kube-ovn	kube-router	macvlan	weave
almalinux8	✅	❌	❌	❌	✅	❌	❌	❌
amazon	✅	❌	❌	❌	❌	❌	❌	❌
centos7	✅	❌	❌	✅	❌	✅	❌	✅
debian10	✅	✅	❌	❌	❌	❌	✅	❌
debian11	✅	❌	✅	❌	❌	❌	❌	❌
debian12	✅	✅	❌	❌	❌	❌	❌	❌
fedora37	✅	❌	❌	❌	❌	✅	❌	❌
fedora38	❌	❌	❌	❌	✅	❌	❌	❌
opensuse	❌	❌	❌	❌	❌	❌	❌	❌
rockylinux8	✅	❌	❌	❌	❌	❌	❌	❌
rockylinux9	✅	✅	❌	❌	❌	❌	❌	❌
ubuntu20	✅	✅	❌	✅	❌	✅	❌	✅
ubuntu22	✅	❌	❌	❌	❌	❌	❌	❌

EQUINIX

Thanks for Sponsor

Best Practice: NTP & Sysctl

NTP

Time synchronization is very important for the Etcd and kubernetes.

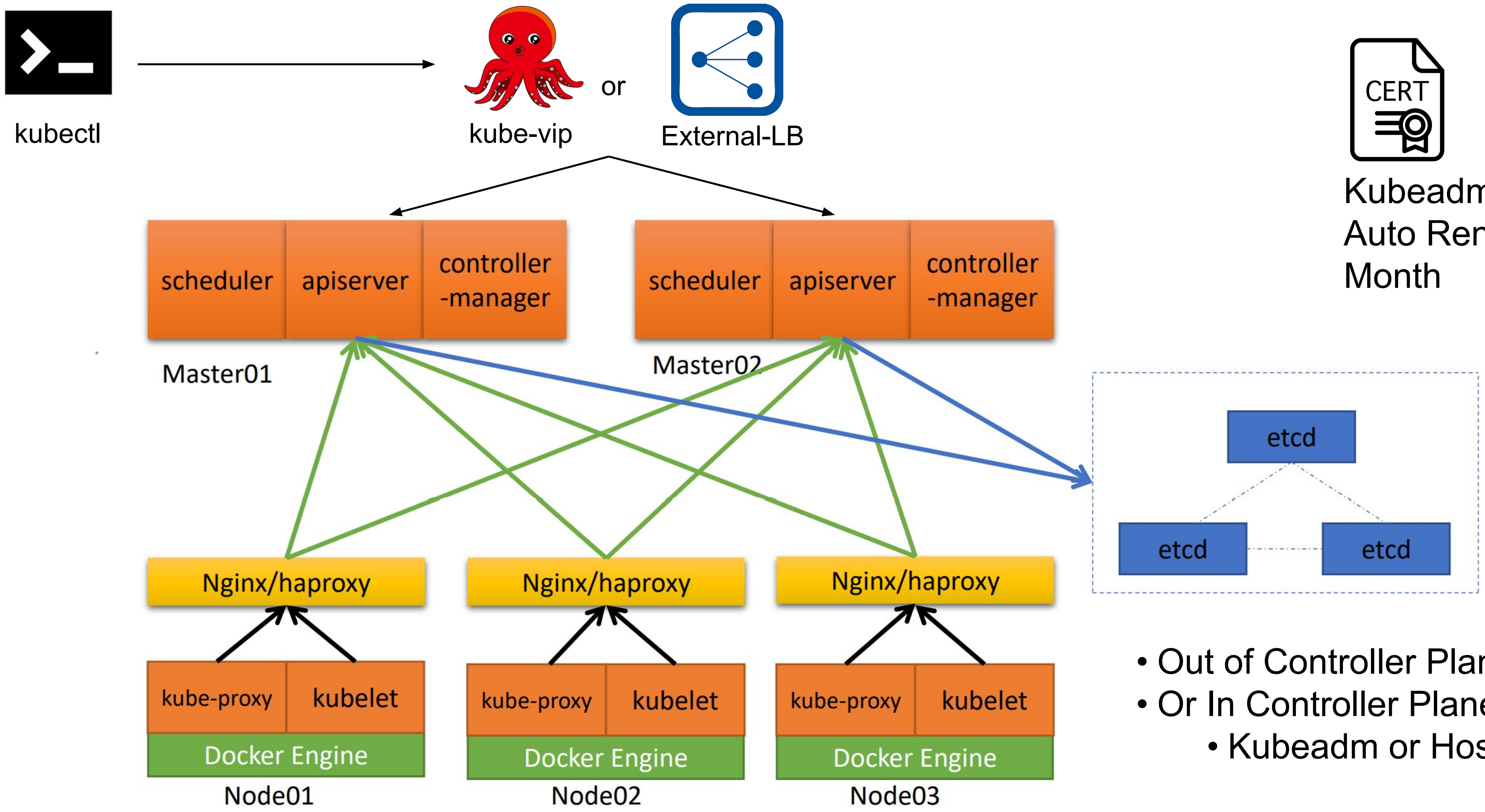
```
ntp_enabled: true
ntp_timezone: Asia/Shanghai
ntp_manage_config: true
ntp_force_sync_immediately: true
```

Sysctl

- Required **kernel variables** has been configured by default.
- Support customize the additional variables for tunning

```
additional_sysctl:
 - { name: kernel.pid_max, value: 4194304 }
 - { name: net.netfilter.nf_conntrack_max, value: 1048576 }
 - { name: fs.inotify.max_user_watches, value: 65536 }
 - { name: fs.inotify.max_user_instances, value: 8192 }
```

Best Practice: High Availability



CERT

Kubeadm Cert Auto Renew Every Month

- Out of Controller Plane Node
- Or In Controller Plane Node
  - Kubeadm or Host Mode

Best Practice: Ansible Collection

Kubespray can work well with Ansible Collection.

1.Add Kubespray to your requirements.yml file

```
collections:
- name: https://github.com/kubernetes-sigs/kubespray
  type: git
  version: master # use the appropriate tag or branch for the version you need
```

2.Install your collection

```
ansible-galaxy install -r requirements.yml
```

3.Create a playbook to install your Kubernetes cluster

```
- name: Install Kubernetes
  ansible.builtin.import_playbook: kubernetes_sigs.kubespray.cluster
```

4.Update INVENTORY and PLAYBOOK so that they point to your inventory file and the playbook you created above, and then install Kubespray

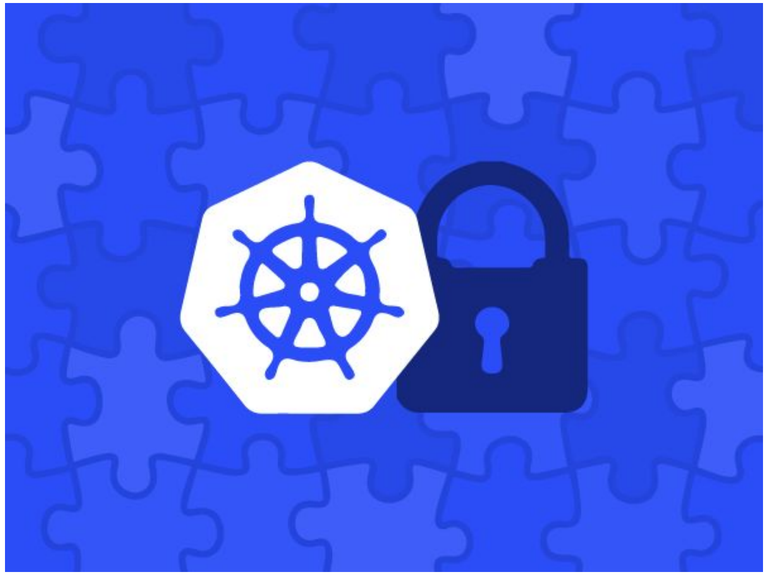
```
ansible-playbook -i INVENTORY --become --become-user=root PLAYBOOK
```

Best Practice: Cluster Hardening

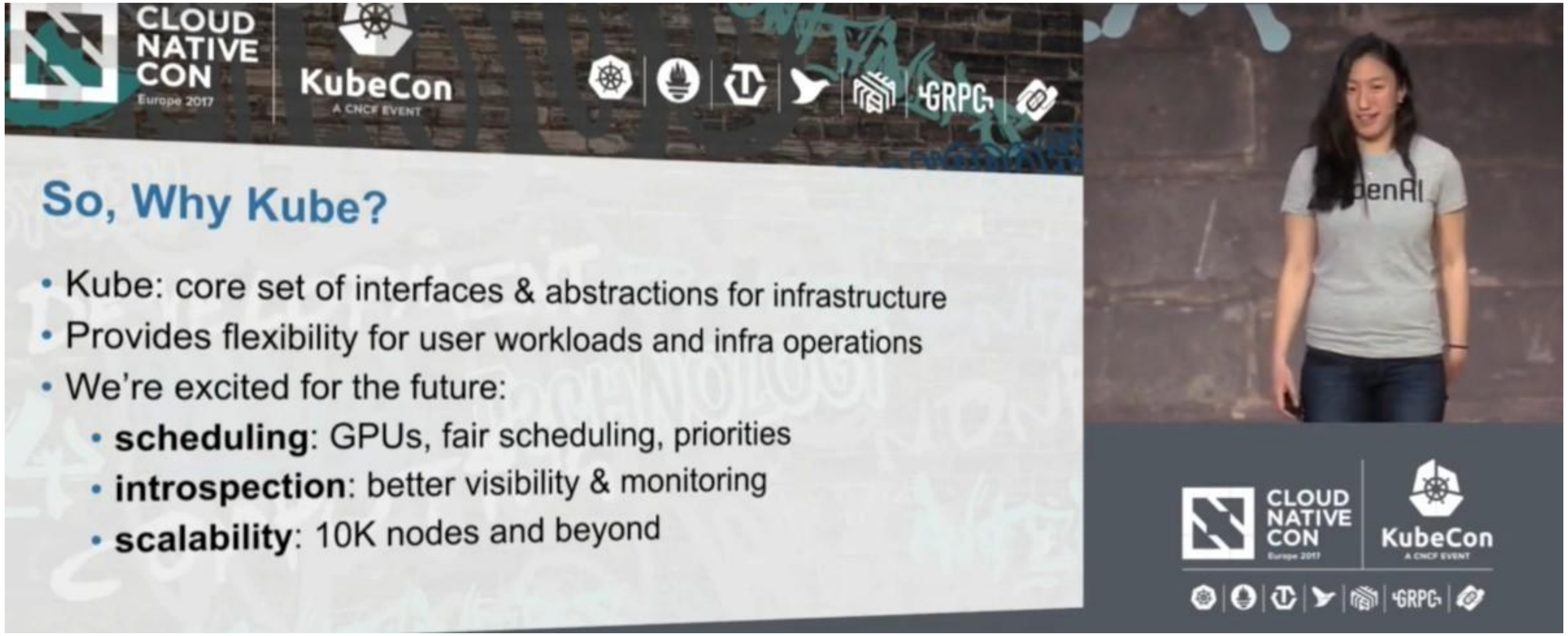
- Full support with config
- Cluster hardening Guide

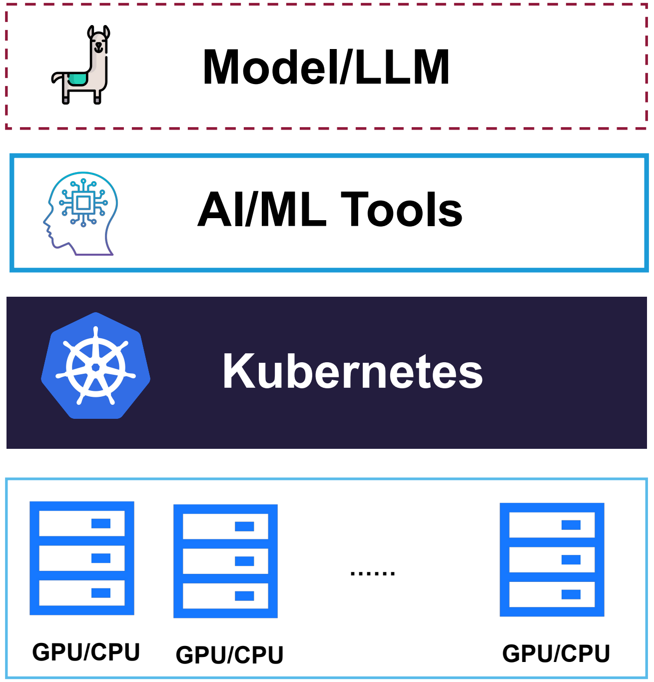
```
## kube-apiserver
authorization_modes: ['Node', 'RBAC']
kube_apiserver_request_timeout: 120s
kube_apiserver_service_account_lookup: true

# enable kubernetes audit
kubernetes_audit: true
audit_log_path:
"/var/log/kube-apiserver-log.json"
audit_log_maxage: 30
audit_log_maxbackups: 10
audit_log_maxsize: 100
......
```

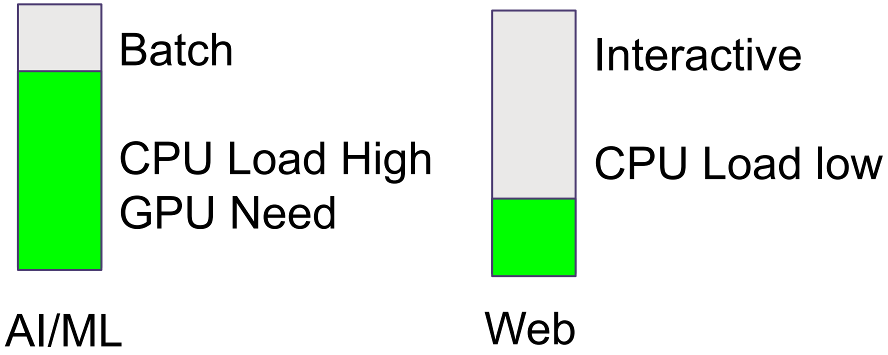


AI/ML Workload for Kubernetes





For Example:
Kubernetes that powers OpenAI infrastructure



Challenges for Kubespray

GPU Support

- Support Device Plugin
- Install Driver From Binary
- Not using Official Nvidia Driver Image
- Lack of Prometheus Exporter
- Lack of Node Feature Discover
- Lack of MIG
- Lack of RDMA

Challenges

Scheduler & Queue

- Kube-Scheduler
- Lack of gang scheduling
- Lack of capacity scheduling
- Lack of priorities scheduling

AI tools

- Lack of AI “App”

AI-Optimized k8s Cluster

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Models

Model Develop | Model Training | Model Serving | LLM

AI Tools

AI/ML Framework
PyTorch | Tensorflow | Hugging face

AI/ML Operators
KubeRay | KubeFlow

Kubernetes

Scheduler for AI/Batch Job
Scheduler Plugin | Volcano

Queue
Kueue

Nvidia GPU Integration
NFD DM RDMA MIG vGPU Kubevirt DRA

Kubernetes

Docker/Containerd/CRI-O Support Nvidia GPU Driver

Infrastructure

Bare Metal

CPU
Intel ARM
AMD

NVIDIA GPU
4090 A100
H100 T4

Cloud

Terraform GPU instance AWS EQUINIX Google Cloud

developing

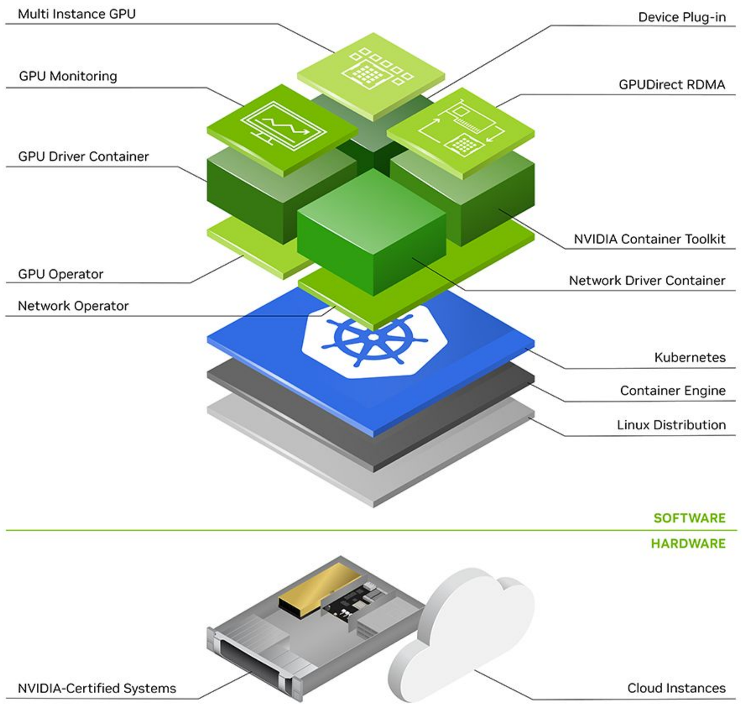
Nvidia GPU Integration

- GPU Operator
  - nvidia-container-toolkit
  - nvidia-gpu-driver

Host-level Components

  - k8s-device-plugin
  - gpu-feature-discovery
  - nvidia-mig-manager
  - dcgm-exporter

Kubernetes Components



Ref to: Unlocking the Full Potential of GPUs for AI Workloads on Kubernetes - Kevin Klues, NVIDIA

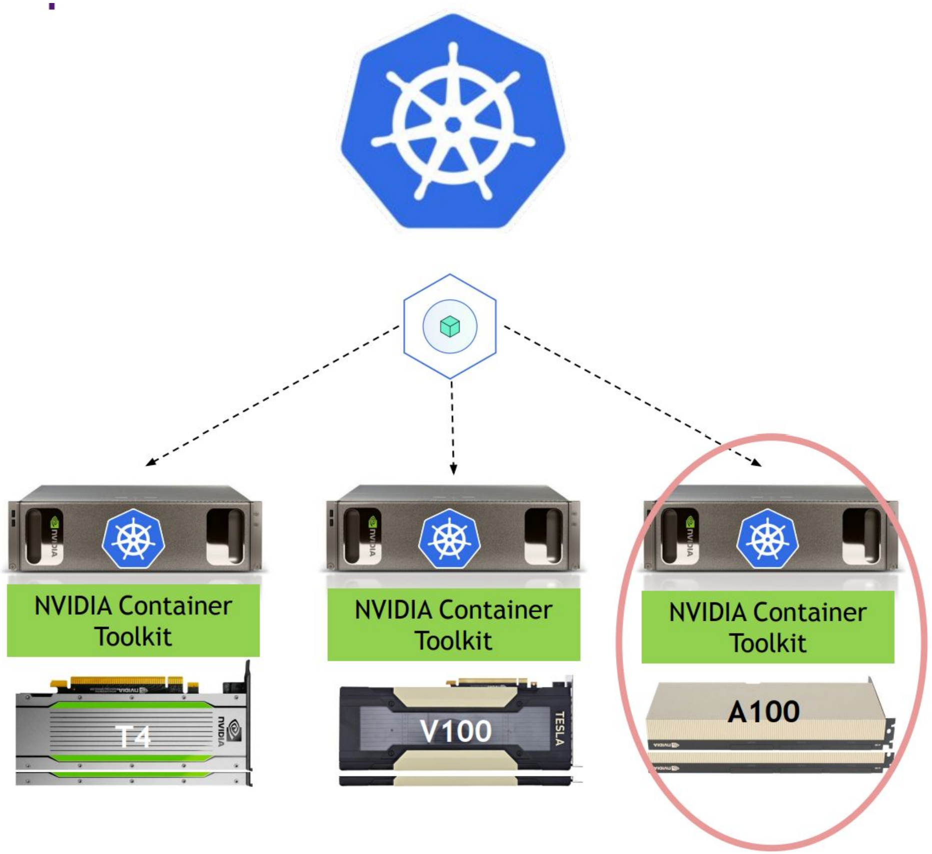
GPU Basic Usage


```yaml
apiVersion: v1
kind: Pod
metadata:
  name: gpu-example
spec:
  containers:
  - name: gpu-example
    image: nvidia/cuda
    resources:
      limits:
        nvidia.com/gpu: 2
  nodeSelector:
    nvidia.com/gpu.product: A100-PCIE-40GB
```

CUDA Image

Schedule by Device Plugin

Label by gpu feature discovery



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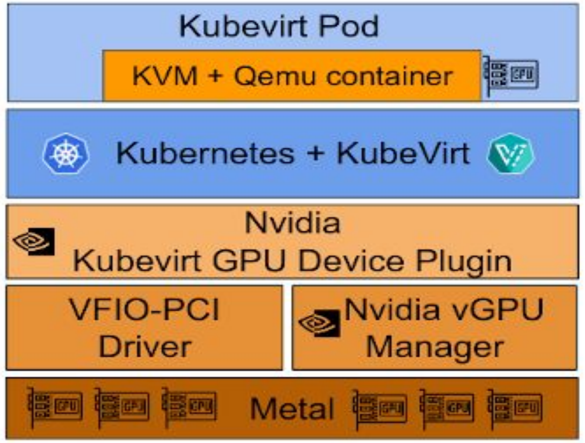
GPU Advanced Usage

MIG

- 1 x 7g.40gb
  or
- 2 x 3g.20gb
  or
- 3 x 2g.10gb
  or
- 7 x 1g.5gb

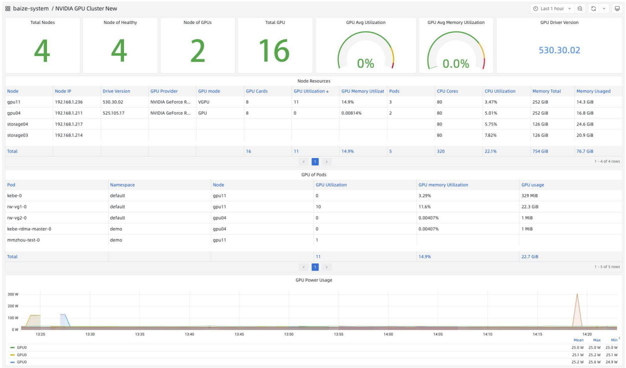
```yaml
apiVersion: v1
kind: Pod
metadata:
  name: gpu-example
spec:
  containers:
  - name: gpu-example
    image: nvidia/cuda
    resources:
      limits:
        nvidia.com/mig-1g.5gb.shared: 1
        nvidia.com/mig-1g.5gb: 1
```

KubeVirt




```yaml
apiVersion: kubevirt.io/v1alpha3
kind: VirtualMachineInstance
metadata:
  name: vmi-gpu
spec:
  domain:
    devices:
      gpus:
      - deviceName:
nvidia.com/GP102GL_Tesla_P40
        name: gpu1
...
```

DCGM exporter



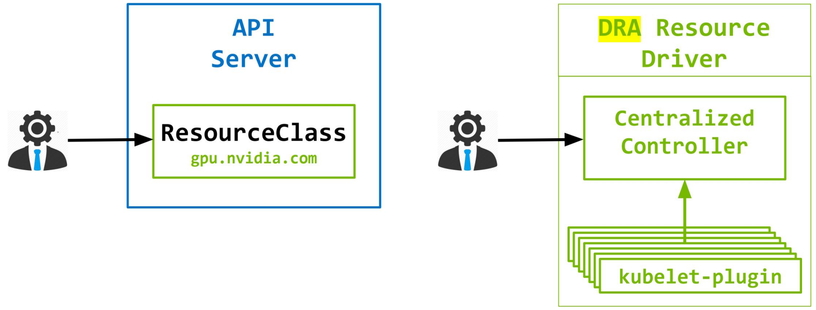
Grafana Prometheus NVIDIA DCGM

ref: https://docs.nvidia.com/datacenter/tesla/mig-user-guide/index.html

GPU Integration Future work

Dynamic Resource Allocation (DRA)

- New way of requesting resources available (as an alpha feature) since Kubernetes 1.26+
- An **alternative** to counting-based interface in the Device plugin



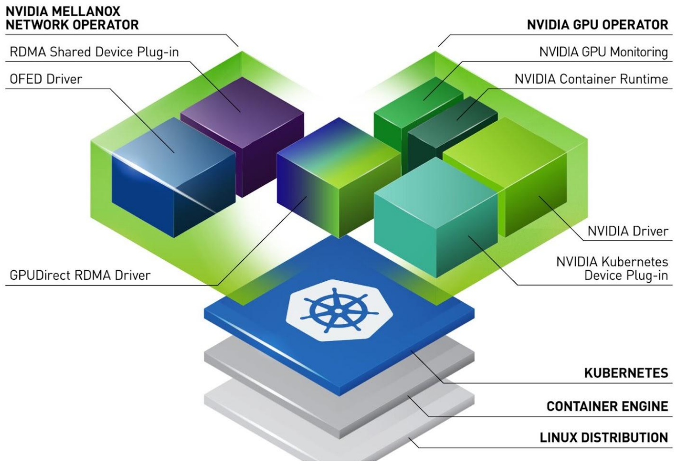
Mellanox network-operator

- GPUDirect RDMA

MacVlan NIC

```
apiVersion: v1
kind: Pod
metadata:
  name: demo-pod-1
  annotations:
    k8s.v1.cni.cncf.io/networks:
demo-macvlannetwork
spec:
  containers:
  - image: mellanox/cuda-perftest
    name: rdma-gpu-test-ctr
    resources:
      limits:
        nvidia.com/gpu: 1
        rdma/rdma_shared_device_a: 1
```

RDMA Device



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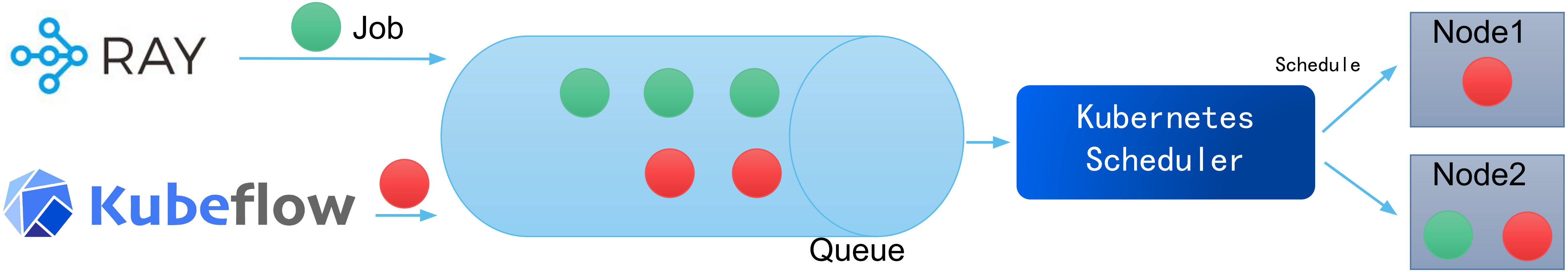
Why Schedule Important for AI/ML



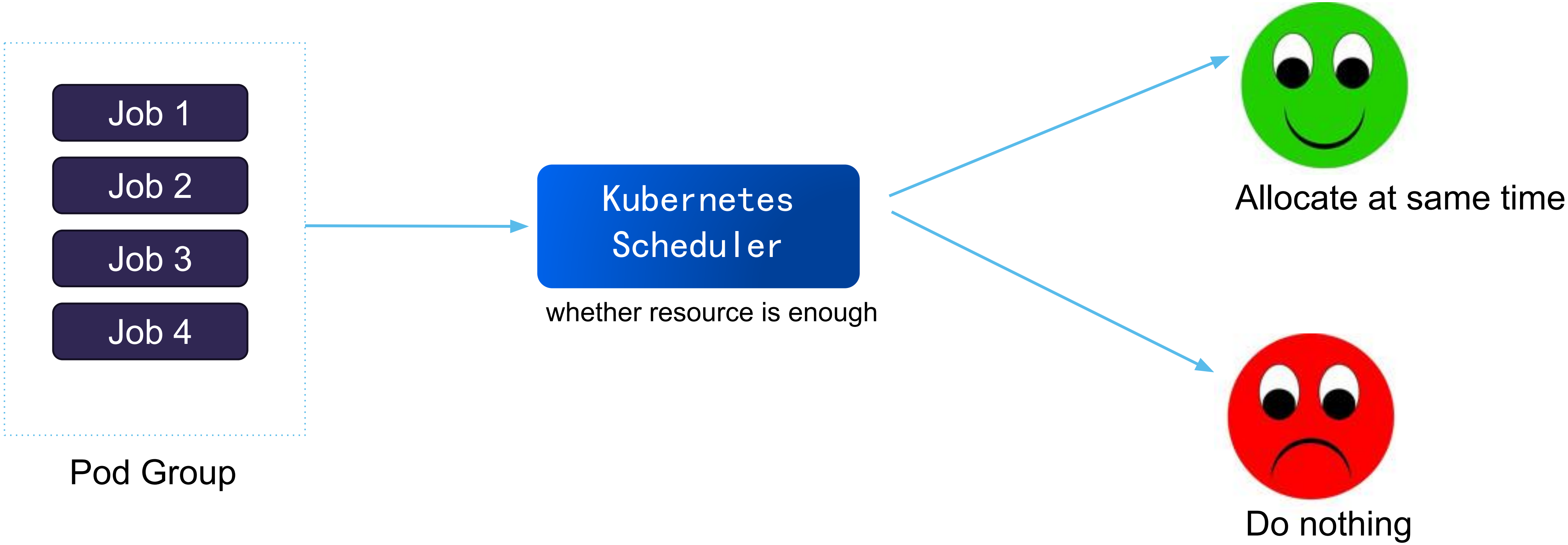
Resource Management:
AI/ML workloads often require large amounts of computational resources. Kubernetes scheduler ensures efficiently utilize available resources.

Distributed Jobs:
Gang scheduling ensures synchronized, efficient execution of distributed AI/ML workloads in Kubernetes, optimizing resource use and preventing computational deadlocks.

Prioritization:
n a multi-tenant environment. Kubernetes scheduling can prioritize critical workloads to ensure they get the necessary resources first.

Gang Scheduling

all pods **must** be up and running at same time



Scheduler for AI/ML Job

Scheduler-Plugin K8S SIG

Incorporates best practices and utilities to compose a high-quality out-tree scheduler plugins based on scheduling framework

```yaml
apiVersion: scheduling.x-k8s.io/v1alpha1
kind: PodGroup
metadata:
  name: nginx
spec:
  scheduleTimeoutSeconds: 10
  minMember: 3
---
apiVersion: apps/v1
kind: ReplicaSet
metadata:
  name: nginx
spec:
  schedulerName: scheduler-plugins-scheduler
  replicas: 6
  selector:
    matchLabels:
      app: nginx
  template:
    metadata:
      name: nginx
      labels:
        app: nginx
        scheduling.x-k8s.io/pod-group: nginx
```

Using Scheduler Plugin

Using PodGroup

Various Scheduling Policy supported

Gang
- avoid waste/ deadlock
- Big jobs starve smalls

Preempt
- Diverse SLO

Binpack
- Avoid fragments
- Small jobs starve bigs

Affinity
- inner-node performance first

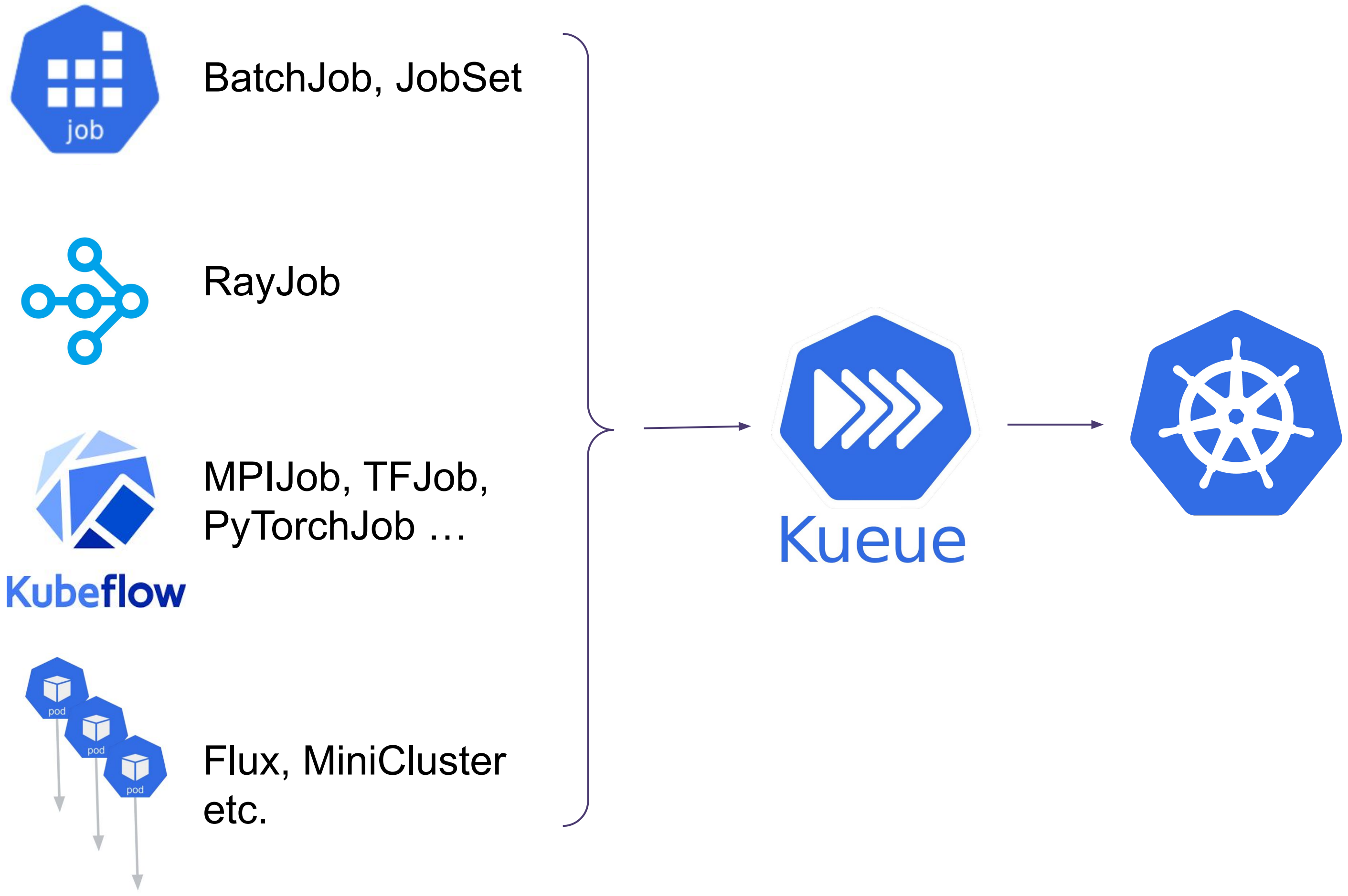
VOLCANO another good choice

Queue for AI/ML Job

Kueue K8S SIG

A Kubernetes-Native job queueing system,

- Resource quota management
  - Two-level hierarchy for fair sharing and borrowing
  - Priority-based ordering and preemption.
  - Fungibility: burst to on-demand, spot, other models.
- Support for k8s batch/v1.Job, JobSet, kubeflow training,
  - RayJob and plain Pods.
- Extensions for:
  - Custom job CRDs.
  - Additional admission checks



Integrated with Job Framework

Ref to: SIG Scheduling Updates KubeCon 2023 US - Kensei Nakada, Aldo Culquicondor

Discuss: Helm vs Ansible Template

Q: Using helm template or ansible template in Kubespray ？

Benefit

- Maintenance by the original project itself

Challenge

- Almost all features is implement by ansible, Migrate means break change

Benefit

- All Templates is include by Kubespray, make it's easy to customized.
- Ready for air-gap environment

Challenge

- huge maintenance load

A: The Kubespray should the helm and ansible template both. Ansible is better for System Components, and Helm is better for Apps.

Demo

https://asciinema.org/a/646990

Thanks to Community

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a Pure Bazaar Global Community

Developers World Wide

1338 Developers

>50 in a release

7536 Commits

VannTen

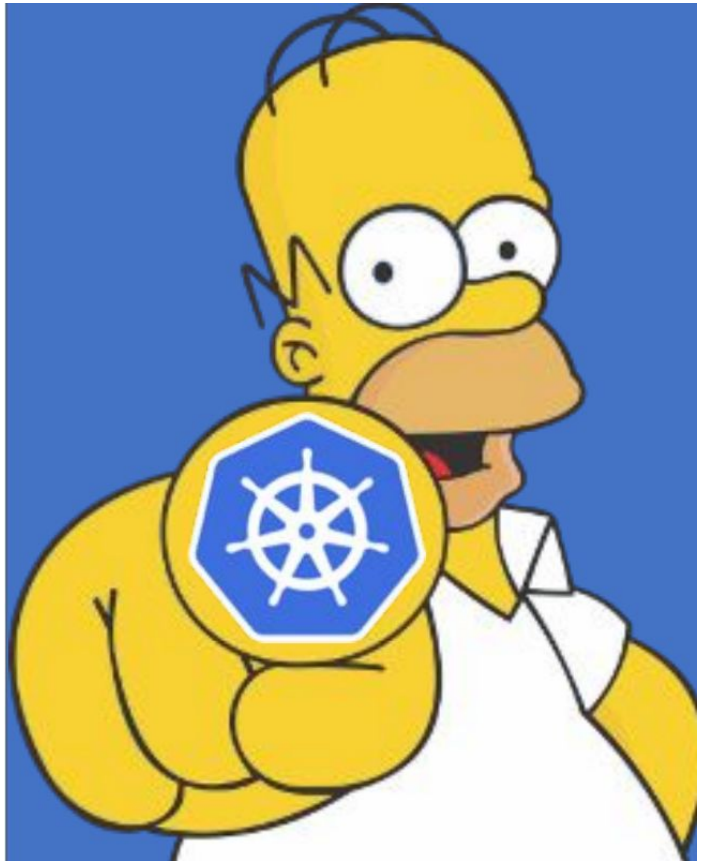
ErikJiang

MrFreezeex

cyclinder

New Maintainers last year

Need your help!



un pour tous, tous pour un.
(EN: One for all, all for one)

- Alexandre Dumas, France

#kubespray for general questions & support
#kubespray-dev for Kubespray development

kubernetes-sigs/kubespray

Kiosk

Questions and Answers